

A New IoT Platform Based on PQC Algorithms: A Solution for Quantum Computing Threats

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Abstract—The migration from classical public-key cryptography to post-quantum cryptography is becoming critical for Internet of Things authentication, because future quantum computers can undermine RSA, Diffie-Hellman, and elliptic-curve mechanisms used in many passwordless systems. This paper proposes a scalable IoT platform that maps post-quantum algorithms to the hardware tier where they are most practical. The proposed architecture uses ML-KEM for lightweight session-key establishment at IoT nodes, ML-DSA for frequent challenge-response authentication, and SLH-DSA for high-assurance but less frequent gateway or server-side attestation. The platform also defines a passkey metadata structure, a hardware data path with DMA-assisted payload movement, and a benchmark methodology on QEMU-based ARM64 and RISC-V64 targets. Measurements from the prototype show that ML-KEM and ML-DSA remain practical for constrained nodes, while SLH-DSA should be isolated to gateway or server tasks due to its large signature size and high signing latency. The result is a migration-oriented design for quantum-resistant IoT authentication without abandoning existing WebAuthn and embedded-system workflows.

Keywords—Post-Quantum Cryptography (PQC), Internet of Things (IoT), Passkey, WebAuthn, ML-KEM, ML-DSA, SLH-DSA, RISC-V

I. INTRODUCTION

The rapid emergence of quantum computing poses an unprecedented threat to the security foundations of the Internet of Things (IoT), because RSA, Diffie-Hellman, and elliptic-curve mechanisms rely on integer factorization or discrete-logarithm assumptions that are vulnerable to Shor's algorithm [1]. Prior hardware and embedded studies show that PQC deployment cannot be treated as a direct drop-in replacement for classical public-key primitives. Configurable lattice crypto-processors and IoT-oriented PQC processors have demonstrated that polynomial arithmetic, sampling, and memory organization dominate the cost of practical lattice-based cryptography [2] - [3]. More recent FPGA and software-hardware co-design work further confirms that Kyber/ML-KEM and Dilithium/ML-DSA benefit from dedicated acceleration and careful data movement [4] - [5], while hash-oriented accelerators are important for SPHINCS+/SLH-DSA-style signatures [6]. At the system level, IoT studies report that energy, bandwidth, and communication cost must be evaluated together rather than as isolated runtime values [7]-[9].

For reproducible implementation and migration experiments, software ecosystems such as PQCclean and the Open Quantum Safe project provide clean C baselines and protocol-integration paths for post-quantum algorithms [10] - [11]. NIST has finalized the first set of PQC standards: ML-

KEM, derived from Kyber, for key encapsulation [12], ML-DSA, derived from CRYSTALS-Dilithium, for digital signatures [13], and SLH-DSA, derived from SPHINCS+, for stateless hash-based signatures [14]. In parallel, WebAuthn defines a scoped public-key credential workflow that can be preserved while the algorithms inside a credential are replaced or hybridized during migration [15]. Despite these advances, existing frameworks still provide limited guidance on how to balance heavy cryptographic demands with heterogeneous IoT device capabilities. To overcome this limitation, this paper proposes a scalable Edge-IoT architecture that maps PQC primitives to the hardware tier where they are most practical and integrates them with passkey-style metadata.

This paper is shown 5 sections, the related works of current resesarches are briefly sumerized in the next section. The third section will be proposed the new architecture of IoT Platform based on PQC Algorithms. The fourth will show some experiment results of IoT Platform based on PQC Algorithms in term of connectivy and authentication speed. The final is conclusion section.

II. RELATED WORKS

A. NIST PQC Standards

ML-KEM, formerly CRYSTALS-Kyber, is a key-encapsulation mechanism based on module learning with errors. It provides a shared secret and ciphertext, but it is not intended to encrypt application data directly. A practical protocol uses ML-KEM to establish a session secret and then applies a symmetric cipher such as AES or ChaCha20 to the actual IoT payload. ML-DSA, formerly CRYSTALS-Dilithium, provides digital signatures based on module-lattice problems and is suitable for frequent authentication. SLH-DSA, derived from SPHINCS+, is hash-based and conservative, but its signatures are much larger and its signing procedure is slower than lattice-based alternatives [12]-[14]. Open-source libraries also influence deployability. PQCclean focuses on clean, portable, tested C implementations of post-quantum algorithms [10], whereas the Open Quantum Safe project provides a higher-level software ecosystem for prototyping quantum-safe cryptography and integrating algorithms into Internet protocols [11]. This paper assumes reference-style C implementations during measurement because they provide a reproducible baseline before hardware acceleration.

B. WebAuthn, Passkeys, and Quantum migration

WebAuthn defines an API for creating and using scoped, attested, public-key credentials. A credential is created for a

WebAuthn relying party, stored by an authenticator, and later used only for origins that belong to the same relying party. Authentication follows a challenge-response model in which the authenticator proves possession of the private key by producing a digital signature over data derived from the server challenge [15]. A post-quantum passkey can preserve this high-level workflow while replacing or hybridizing the signature algorithm inside the credential object. In this paper, the proposed PQC credential is therefore treated as a migration-oriented profile rather than as a claim that current WebAuthn deployments universally support PQC credentials.

A complete PQC credential requires more metadata than a classical credential. The credential must carry the algorithm identifier, public key encoding, parameter set, key usage, format, attestation evidence, user handle, credential identifier, and optional hybrid-mode indicator. These fields are necessary because ML-DSA and related PQC algorithms expose multiple parameter sets, larger public keys and signatures, and specific COSE or JOSE representations [16]. They also allow the relying-party server and gateway to distinguish classical-only, hybrid, and PQC-only credentials during a transition period. Draft-level work on ML-DSA for WebAuthn further shows that backward compatibility, large signature handling, authenticator behavior, and COSE algorithm support must be considered explicitly when integrating PQC signatures into passwordless authentication [17].

C. Embedded communication and boot-chain context

The embedded platform around the cryptographic library is equally important. UART is simple and useful for logging or command channels, but it is inefficient for large PQC payloads. SPI, SDIO, USB, and eMMC provide higher throughput and may support DMA transfers. On RISC-V Linux systems, the boot chain also involves firmware layers such as OpenSBI and U-Boot before the kernel starts. A practical PQC platform should therefore connect cryptographic policy, storage, transport, and boot-time trust rather than treating PQC as a single isolated library call.

D. Migration gap in current IoT stacks

Most existing IoT security stacks were built around compact classical primitives. Replacing ECDSA or ECDH with PQC changes buffer sizes, packet fragmentation behavior, nonvolatile storage layouts, and the cost of certificate or credential exchange. These changes become visible at the application level when the signed object is larger than the original telemetry record. A practical design therefore needs explicit payload policies rather than a blind algorithm substitution.

The migration gap is also operational. During a transition period, some devices will support only classical credentials, some will support hybrid credentials, and new devices may support PQC-only credentials. The platform proposed in this paper treats hybrid mode as a policy state. This avoids forcing a single cutover point and allows a gateway to mediate between legacy and PQC-capable devices while the server maintains a unified credential registry.

III. PROPOSED ARCHITECTURE

A. Layered deployment model

Fig. 1 shows the proposed architecture. The IoT node owns the device-bound credential and performs the operations that are latency-critical for local communication. The gateway acts as a security concentrator that translates protocols, aggregates signatures, applies rate limits, and offloads heavier attestation operations. The relying-party server stores the public credential metadata, verifies authentication results, and manages key lifecycle events such as revocation, rotation, and policy updates.

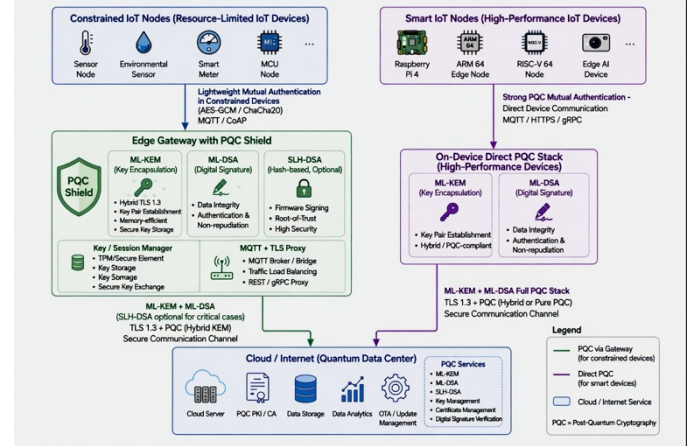


Fig. 1. Proposed Architecture of IoT Platform based on PQC Algorithm.

The platform separates three cryptographic functions. ML-KEM is assigned to node-level session establishment because its ciphertexts and shared-secret outputs are compact enough for embedded communication. ML-DSA is used when the node must sign a challenge or bind a session to a device identity. SLH-DSA is reserved for firmware, audit, and long-term attestation flows because its hash-based security margin is attractive but its large signatures are not efficient for high-frequency sensor traffic.

Table I. PQC algorithm placement in the proposed platform

Primitive	Placement	Design rationale
ML-KEM	End node (IoT node)	Session-key establishment with compact ciphertext and shared secret.
ML-DSA	End Node or Gateway	Frequent challenge-response authentication with moderate signature size.
SLH-DSA	Gateway or Server	Conservative hash-based attestation, but slow and signature-heavy.
Hybrid mode	All tiers	Transition path for devices that still require classical compatibility.

B. PQC passkey object and policy

The proposed credential object extends the passkey model with PQC-specific metadata. The minimum fields are alg, publicKey, parameters, keyUsage, format, attestation, userHandle, and credentialId. The hybrid field indicates whether the credential is bound to both a classical algorithm and a PQC algorithm. The parameters field stores the security level and parameter set, for example ML-KEM-512 or ML-DSA-65, so that the verifier does not misinterpret a public key or signature buffer.

The policy engine selects the algorithm according to message type. Short-lived telemetry sessions use ML-KEM plus symmetric encryption. Login and device enrollment use ML-DSA signatures. Firmware release signing, gateway audit records, and root-of-trust evidence may use SLH-DSA because those events are less frequent and can tolerate larger payloads. This policy reduces pressure on constrained devices while still retaining a conservative security option for long-lived records.

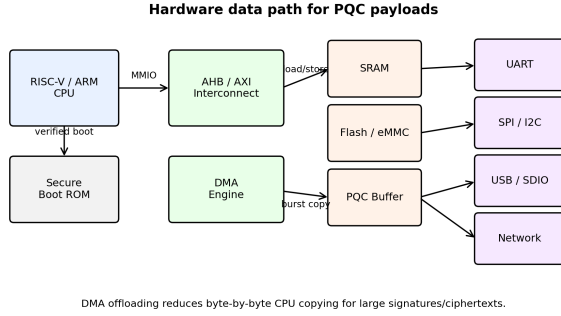


Fig. 2. Hardware Architecture of End Node using PQC Algorithm in IoT Platform.

C. Hardware data path

Fig. 2 illustrates the embedded data path. The CPU controls peripherals through memory-mapped registers, but large PQC objects are moved through shared buffers and DMA where available. This is important because a signature may be several kilobytes or tens of kilobytes, and copying it byte by byte through programmed I/O wastes CPU time and increases response latency. The same concept applies to USB gadget links, SDIO-based Wi-Fi modules, eMMC logging, and SPI-connected secure elements.

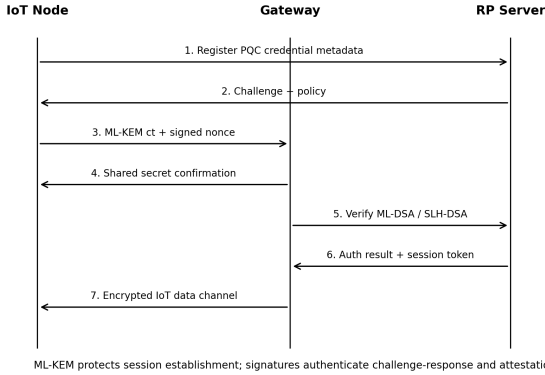


Fig. 3. PQC authentication and session-establishment flow for the proposed IoT platform.

The architecture is compatible with both microcontroller nodes and Linux-capable boards. On a small microcontroller, the implementation can be reduced to a static credential object, a KEM module, and a transport interface. On a Linux-capable RISC-V or ARM board, the platform can use a boot chain, file system, and network stack to separate attestation, logging, and policy enforcement. This makes the design scalable from ESP32-class devices to edge gateways.

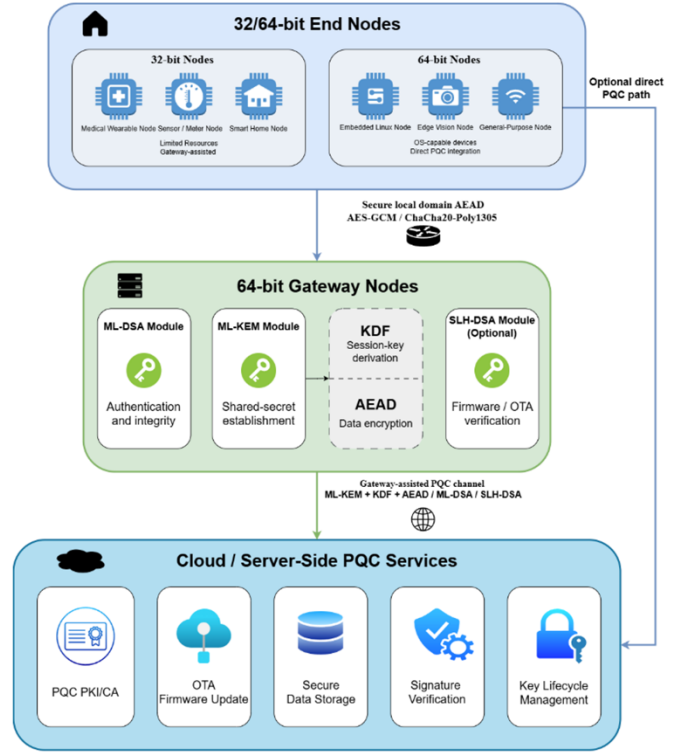


Fig. 4. Quantum-safe session establishment and passkey authentication flow.

D. Enrollment, authentication, and key rotation

The platform separates credential enrollment from runtime authentication. During enrollment, the device creates or receives a PQC-capable credential profile, including the algorithm identifier, public key, parameter set, and attestation metadata. The relying party stores only the public credential and policy state. During authentication, the server sends a nonce and a policy challenge. The node or gateway signs the challenge, and ML-KEM can be used in the same session to establish a fresh symmetric key for application data.

Key rotation is handled at the policy layer rather than by hardcoding a single algorithm. When a parameter set is deprecated or a device class becomes too weak, the server can require re-enrollment with a stronger profile. Gateway-mediated rotation is useful when sleeping nodes cannot maintain continuous network contact. In this mode, the gateway buffers policy updates and applies them when the node wakes, reducing battery and radio overhead.

E. Security boundaries and failure handling

The node boundary protects device-bound secrets and performs the minimum cryptographic work needed to authenticate itself. The gateway boundary protects aggregation state, rate limits repeated verification attempts, and prevents a single slow signature operation from blocking all sensor traffic. The server boundary protects long-term policy, audit records, and credential revocation. These boundaries make the system resilient to partial compromise because a low-end node does not need to hold every signing key or perform every attestation operation.

Failure handling is deliberately conservative. If a node cannot process a required PQC profile, the server can downgrade only to a policy-approved hybrid path and record the event. If a gateway is overloaded, the system can accept telemetry encrypted with a symmetric key established through ML-KEM but delay expensive attestation verification. If a SLH-DSA signature is required for firmware or audit records, the platform does not replace it with a lighter algorithm silently; it schedules the operation at a tier that can handle the cost. This prevents performance optimizations from weakening the security objective.

IV. EXPERIMENT AND RESULTS

A. Experimental setup

The prototype evaluation used C implementations of Kyber/ML-KEM, Dilithium/ML-DSA, and SPHINCS+/SLH-DSA. The single-operation experiments were executed on Ubuntu Server images in QEMU-based virtual ARM64 and RISC-V targets. The QEMU configuration used one virtual core and 296 MB RAM, with NTESTS = 10000 for ML-KEM and ML-DSA and NTESTS = 1000 for SLH-DSA because the hash-based signature family is significantly slower. Additional stress tests simulated high-load IoT conditions with approximately 50 or more concurrent message-processing threads.

The measured outputs include average key generation time, signing or encapsulation time, verification or decapsulation time, all-time NIST-style execution time, public key size, secret key size, signature size, ciphertext size, shared secret size, elapsed response time, CPU user time, CPU system time, and maximum resident memory. These metrics were selected because IoT deployment is constrained not only by cryptographic security but also by latency, memory, and link bandwidth.

The experiments were not designed to choose a universal winner. Instead, they were designed to expose which operation dominates a given deployment tier. For key establishment, the critical values are ciphertext size, decapsulation time, and shared-secret availability. For signatures, the critical values are signing time, verification time, signature length, and the number of concurrent verification requests that a gateway can absorb. This tier-oriented view is more useful for IoT architecture than a single average runtime number.

B. Single-operation observations

The QEMU results support the proposed placement policy. On the tested ARM64 target, ML-KEM end-to-end operation time was approximately 1.00 ms for Kyber512, 1.40 ms for Kyber768, and 2.02 ms for Kyber1024. ML-DSA was heavier but still practical for authentication, with end-to-end values of approximately 4.07 ms for Dilithium2, 6.42 ms for Dilithium3, and 8.41 ms for Dilithium5. On the RISC-V target, the same operations were in a similar order of magnitude, although generally slower for the selected configurations.

SLH-DSA behaved differently. For selected RISC-V 64-bit SPHINCS+ parameters, sha2-128f-simple signed in about 205.06 ms with a 17088-byte signature, while shake-256f-robust signed in about 2508.36 ms with a 49856-byte signature. This confirms that SLH-DSA is a valuable high-assurance option but

not a good default for frequent low-latency IoT messages. The architecture therefore treats SLH-DSA as a gateway/server operation or an infrequent firmware and audit operation.

Table II. Stress workload summary from the prototype measurement

Algorithm	Measured stress metrics (elapsed, RSS, payload)	Recommended tier
Dilithium2	1110.97 ms, 628 KB, 2420 B sig.	Node/gateway
Dilithium3	1406.81 ms, 660 KB, 3309 B sig.	Gateway
Dilithium5	1768.26 ms, 704 KB, 4627 B sig.	Gateway
Kyber512	1589.82 ms, 704 KB, 768 B ct.	Node
Kyber768	1584.53 ms, 644 KB, 1088 B ct.	Node
Kyber1024	1822.05 ms, 652 KB, 1568 B ct.	Node/gateway
SPHINCS+	29463.60 ms, 612 KB, 17088 B sig.	Server

C. Stress-test analysis

Table II and Fig. 4 summarize the stress workload. The signature-based algorithms show a direct payload penalty: Dilithium grows from 2420 to 4627 bytes across the tested security levels, while SPHINCS+ reaches 17088 bytes and approximately 29.46 s elapsed time in the measured stress workload. The Kyber family has smaller ciphertexts, from 768 to 1568 bytes, and low CPU user time compared with the signature-heavy cases. This supports the decision to keep ML-KEM at the node and to avoid per-message SLH-DSA signing at the edge.

The memory column in Table II is a Linux/QEMU resident-set measurement, not a direct SRAM-footprint estimate for an ESP32-class microcontroller. It should therefore be used as a relative software-side indicator rather than as proof that the same workload fits unchanged in microcontroller SRAM. The dominant constraint is the combination of response time, payload length, and concurrency. A node that transmits small sensor messages over a low-bandwidth link can spend more bandwidth on the signature than on the application data. Therefore, the proposed platform batches, aggregates, or offloads heavy signatures at the gateway. For message-level confidentiality, the node establishes a shared secret using ML-KEM and then encrypts the application stream with a symmetric cipher.

These results should be interpreted as prototype measurements rather than final silicon characterization. QEMU is useful for comparing software flows and parameter sets, but final deployment should include power measurement, real radio throughput, flash-wear analysis, and timing leakage tests on physical boards. Nevertheless, the relative behavior is clear enough for system design: lattice-based KEM and DSA are practical building blocks for IoT nodes, while hash-based signatures are best used at lower frequency or at a higher resource tier.

D. Design implications

The first implication is that PQC integration should be buffer-first. Static buffers sized for classical signatures are not enough. The firmware should define separate buffers for KEM ciphertexts, digital signatures, credential metadata, and

application payloads, then enforce maximum sizes at compile time. This avoids accidental stack growth and makes memory pressure visible during review. For RTOS-based nodes, long signatures should be allocated in controlled memory regions rather than on small task stacks.

The second implication is that authentication should be rate-aware. A gateway that verifies many signatures concurrently may become the bottleneck even if each individual verification is acceptable. The gateway therefore needs queues, timeouts, and admission control. When traffic bursts occur, ML-KEM-based encrypted telemetry can continue while noncritical audit signatures are deferred. This is particularly useful in sensor fleets where data freshness and identity assurance have different real-time requirements.

The third implication is that hardware acceleration should target the dominant kernels rather than the protocol wrapper. For lattice-based schemes, polynomial arithmetic and number-theoretic transform units are natural acceleration targets. For hash-based signatures, SHA-2 and SHAKE throughput dominate. A future FPGA or ASIC implementation should therefore expose memory-mapped accelerator registers and DMA descriptors so that the software stack can reuse the same architecture described in Fig. 2.

Finally, testing must include design-for-test and fault-injection considerations. PQC accelerators will be used in security-critical SoCs, so they should support scan, built-in self-test, and fault detection without exposing secrets. The proposed architecture can be extended with a test mode that validates data-path integrity, accelerator control registers, and memory isolation before enabling production credentials.

V. CONCLUSION

This paper proposed a scalable IoT platform for post-quantum authentication and session establishment. The design connects NIST-standardized PQC algorithms with a WebAuthn-style credential model and an embedded hardware data path. The central idea is algorithm placement: ML-KEM is kept close to the node for efficient key establishment, ML-DSA supports frequent challenge-response authentication, and SLH-DSA is used selectively for high-assurance but infrequent attestation.

The prototype measurements show that this division is practical. ML-KEM and ML-DSA have manageable latency and payload sizes for constrained systems, whereas SLH-DSA provides a conservative hash-based fallback at the cost of much larger signatures and higher signing latency. The stress-test analysis further shows that payload size and concurrency can dominate the design even when peak memory is acceptable. Future work will implement the architecture on real ESP32 and RISC-V boards, add NTT and SHAKE/SHA-2 hardware accelerators on FPGA or ASIC, and integrate design-for-test mechanisms for reliable cryptographic SoC deployment.

The main limitation of the present version is that the evaluation is software-centric and simulation-assisted. A full conference submission should add physical-board power results, larger concurrent workloads, and a comparison with at least one existing PQC IoT architecture. Even so, the current draft provides a coherent baseline that connects algorithm choice,

passkey metadata, hardware interfaces, and measured resource behavior.

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